React – Fundamentals

Session - 1

1. Install Node.js and use npx create-react-app to set up a new React project called InstaCloneStarter on your computer.

Ans.

Open a terminal and run:

```
npx create-react-app InstaCloneStarter
```

Move into the project folder:

```
cd InstaCloneStarter
```

Start the development server:

```
npm start
```

2. Open App.js in your InstaCloneStarter project and change the default text to display 'Welcome to My React Zomato App' inside an <h1> tag.

Ans.

```
import './App.css';
import TrendingSong from './TrendingSong';

function App() {
  return (
    <div className="App">
      <h1>Welcome to My React Zomato App</h1>
      <TrendingSong />
    </div>
  );
}

export default App;
```

3. Create a new functional component called TrendingSong.js that returns an <h2> element with the text 'Trending on Spotify: Calm Down'. Import and use this component inside App.js so it shows below your main heading.

Ans.

```
function TrendingSong() {
```

```
  return (
    <h2>Trending on Spotify: Calm Down</h2>
  );
}

export default TrendingSong;
```

4. Explain in your own words (3-4 lines) how React's Virtual DOM makes updating the UI faster compared to traditional DOM manipulation.

Ans.

React's Virtual DOM is a lightweight copy of the real DOM. When the application's state changes, React updates the Virtual DOM first and compares it with the previous version to identify only the changed elements. It then updates only those specific parts of the real DOM instead of re-rendering the entire page. This reduces unnecessary DOM operations and improves UI performance.